

Adolescent and adult soy food intake and breast cancer risk: results from the Shanghai Women's Health Study

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Background: Soy food is a rich source of isoflavones—a class of phytoestrogens that has both antiestrogenic and anticarcinogenic properties. Objective: The objective was to evaluate the association of adolescent and adult soy food intake with breast cancer risk in a cohort of 73,223 Chinese women who participated in the Shanghai Women's Health Study. Design: A validated food-frequency questionnaire was used to assess usual dietary intake during adulthood and adolescence. After a mean follow-up of 7.4 y, 592 incident cases of breast cancer were identified for longitudinal analyses by using Cox regressions. Results: Adult soy food consumption, measured either by soy protein or isoflavone intake, was inversely associated with the risk of premenopausal breast cancer, and the association was highly statistically significant (P for trend < 0.001). The multivariate-adjusted relative risks (RRs) for the upper intake quintile compared with the lowest quintile were 0.41 (95% CI: 0.25, 0.70) for soy protein intake and 0.44 (95% CI: 0.26, 0.73) for isoflavone intake. High intake of soy foods during adolescence was also associated with a reduced risk of premenopausal breast cancer (RR: 0.57; 95% CI: 0.34, 0.97). Women who consumed a high amount of soy foods consistently during adolescence and adulthood had a substantially reduced risk of breast cancer. No significant association with soy food consumption was found for postmenopausal breast cancer. Conclusion: This large, population-based, prospective cohort study provides strong evidence of a protective effect of soy food intake against premenopausal breast cancer.